

SOHEL RAHMAN

Software Engineer

www.srahman.io | github.com/srahman-057 | sohelm.rahman97@gmail.com | (516) 851-9240

Full stack engineer with an interest in machine learning, cloud engineering and game dev. Experienced in building fast, scalable applications that operate in a variety of web or desktop settings.

QUALIFICATIONS

Technical Skills:

Programming: Python, Javascript, Typescript, C, C++, Java

Frameworks: ReactJS, NodeJS, ExpressJS, NextJS, Django

Database: PostgreSQL, MySQL

Languages Spoken: English, Bengali, Hindi

Software System: Git, JIRA

Automation: React Testing Library

Cloud Platforms: AWS, Azure

EXPERIENCE

Barneel Inc. | New York, New York

Founding engineer

(Jan 2025 - Jul 2025)

- Took end-to-end ownership of an experimental project to develop a Django based backend for potential replacement of the firm's existing Node/Express stack.
- Designed scalable RESTful APIs using Django, implementing token-based authentication and role-based permissions for secure data access.
- Built reusable TypeScript interfaces and type definitions to enforce consistent data structures between frontend and backend, reducing risk of integration bugs.
- Managed deployment to AWS using Elastic Beanstalk and utilized Load Balancing to improve application performance and scalability for a multinational target audience.

Web Development Intern

(Oct 2024 - Jan 2025)

- Collaborated with senior developers on production projects, gaining hands-on experience implementing features and fixing bugs in a professional setting.
- Built reusable React components for efficient usage throughout the codebase.
- Contributed to API development using Node.js and Express, integrating with databases and third-party services like Stripe to support application functionality.
- Participated in code reviews and learned industry best practices for version control, testing, and deployment workflows.

Turtle Solutions Ltd. | Dhaka, Bangladesh

(Dec 2019 - Dec 2021)

Software engineer

- Utilized Bootstrap and jQuery to develop interfaces and new features for the company website.
- Spearheaded development of the website's backend from pure PHP to Laravel.
- Migrated website's MySQL database to PostgreSQL, leading to improved reliability and scalability.
- Involved in Enterprise Resource Planning (ERP) software development for a number of clients.
- Received hands-on experience regarding efficient workflow in a professional setting.

PROJECTS

ptch-blog: Full stack web application that offers efficient blog management.

(Sep 2024 - Present)

- Designed modular React components that can be used to deploy dynamically sized storycards at will.
- Implemented Node/ExpressJS backend to handle API routing intuitively and efficiently.
- Optimized production build using Vite.
- Enabled automatic deployments to Vercel based on Github repository updates.
- Used CORS (Cross-Origin Resource Sharing) to prevent unauthorized API access.
- Utilized Arcjet for bot detection, rate-limiting and other security features.

Technology used: NodeJS, ExpressJS, ReactJS, PostgreSQL, Vite, Arcjet

retro_site: Web project that aims to recreate a nostalgic UI look and feel by drawing inspiration from retro operating systems such as Windows 95 and 98. (Jul 2024 - Aug 2024)

- Designed the UI/UX to be accessible from any viewport size (mobile, tablet, desktop, etc).
- Meticulously tested the site to ensure fluid transitions between different viewpoint sizes upon window resizing.
- Optimized the project to minimize AWS storage and bandwidth resource usage and stay within desired quotas.
- Deployed on AWS S3 and utilized a custom domain to make the project easily accessible to the public.
- Implemented a client-side scripting only model in order to further reduce usage of AWS resource quotas.
- Debugged any issue that appeared throughout the development process that interfered with application functionality and efficiency.

Technology used: HTML, CSS, Javascript, Tailwind CSS, AWS S3

genetic-arcade: Python based project that showcases the power of genetic algorithms in a variety of visual environments. (May 2024 - July 2024)

- Built machine learning models that used unsupervised learning to solve diverse problem sets in a changing visual environment.
- Designed code implementing a genetic algorithm (NeuroEvolution of Augmenting Topologies - NEAT) for tackling evolving problem sets.
- Implemented Git version control to effectively document various stages of the project development and keep track of evolving features in the codebase.

Technology used: python, pygame, python-neat, matplotlib

pdf_extractor: Python program that extracts specific data from a table in a pdf file. (May 2023)

- Introduced bash scripting to automate data extraction and organization.
- Included dummy files with mock data to demonstrate usage principles.
- Extensively documented methods to customize the program in order to fit individual scenarios.

Technology used: tabula, pandas, numpy, bash scripting

EDUCATION

Master of Science (M.S.) in Computer Science, The City College of New York (2022-2024)

Bachelor of Science (B.S.) in Computer Science and Engineering, North South University (2016-2021)

HONORS/AWARDS

Full Tuition Waiver Scholarship, North South University, 2016

Kennedy-Lugar Youth Exchange and Study scholarship, Forest Hills High School, New York, 2014